
System Design

Robotic Intervention for Pollution and Plastic Litter Extraction (RIPPLE)

MECHTRON 4TB6: Mechatronics Capstone Design Process

Dr. Alan Wassying

Group 14

December 21st, 2025

Kaelyn Bamford	400299080
Vedashri Pandya	400314055
Shayda Rahmatyan	400322952
Thuvarakaa Sathiyamether	400318002
Clara Yaromich	400311710
Avery Zeiler	400305001

Table of Contents

Table of Contents	II
Table of Tables	III
Table of Figures	III
Purpose of Document	1
Scope of Project	1
Definitions	2
System Diagram Showing Boundaries	3
Monitored & Controlled Variables	4
Constants with Nominal Values & Ranges	5
Behaviour Overview	6
Sub-System 1: Boat Hull	7
Design	7
Boat Hull Overview	7
Inputs & Outputs	9
Description of Behaviour	9
Testing	10
Sub-System 2: Motor Control	11
Motor Control Structural Overview & System Behaviour	11
Inputs & Outputs	11
Design	14
Undesired Event Handling	16
Sub-System 3: MCU	17
Description of Behaviour	17
Schematic	17
Inputs & Outputs	18
Software Design: Initialization	19
GPS	19
Camera	20
Sub-System 4: UI	21
Inputs & Outputs	21
Description of Behaviour	21
Software Design	21
Module Guide	22
Module Decomposition	23
Undesired Event Handling	29
Timing Constraints	29
References	31

Table of Tables

Table 1: Definitions.....	2
Table 2: Monitored Variables	4
Table 3: Controlled Variables	4
Table 4: Constants with Nominal Values & Ranges.....	5
Table 5: Description of Boat Hull Components	8
Table 6: Functional Description of RC Controller Switches	12
Table 7: Functional Description of RC Receiver Outputs.....	13
Table 8: RC Controller Settings and Explanation.....	14
Table 9: MCU System Outputs	18
Table 10: GPS Initialization Parameters	19
Table 11: UI System Inputs	21
Table 12: UI System Outputs.....	21
Table 13: UI System Module Guide	22
Table 14: Undesired Event Handling.....	29
Table 15: Performance Timing Requirements.....	29

Table of Figures

Figure 1: System Diagram Showing Boundaries.....	3
Figure 2: Context Diagram	5
Figure 3: Boat Hull Overview Diagram.....	7
Figure 4: Pool Visual.....	10
Figure 5: Motor Control System Structure.....	11
Figure 6: RC Controller.....	13
Figure 7: Sub-System 3 Electrical Schematic	17
Figure 8: UML Class Diagram.....	23
Figure 9: GPS Module (Left Column) & Camera Module (Right Column)	23

Purpose of Document

The purpose of this document is to detail the design of a remote-controlled surface-water trash collecting system to satisfy defined behavioural and safety requirements. The document will decompose the system into four major sub-systems and describe the inputs/outputs, design details, and desired behaviour of each for a simpler and modular design. Additionally, the document will present how the system handles unexpected events to prevent or mitigate negative consequences of unexpected system errors.

Scope of Project

The system is intended to be used by a trained adult familiar with the controls in a pond or other small body of water that has accumulated trash on its surface in calm, non-precipitating weather. The project is intended to collect the floating trash as directed by its operator.

Definitions

Table 1: Definitions

Word/Acronym	Definition
Boat Hull	This is Sub-System 1. This includes the 3D printed boat and net mechanism.
Motor Control	This is Sub-System 2. This includes the motors, ESC, propellers, and RC.
MCU	This is Sub-System 3. This includes the microcontroller, GPS module, and camera module.
UI	This is Sub-System 4. This is the User Interface.
System	This is the entire system including all Sub-Systems.
TCM	Trash Collection Mechanism. This includes the boat hull and all parts and components attached to it. (All components moving with the boat hull)
RC	Remote/Radio Controller
RID	Requirement Identifier Number
SSR	Safety & Security Requirement
Trash	Floating trash on the surface of the pond that is within the size constraints of the TCM.
ESC	Electronic Speed Controller
HID	Hazard Identifier Number
UM	User's Manual
RPM	Revolutions Per Minute

System Diagram Showing Boundaries

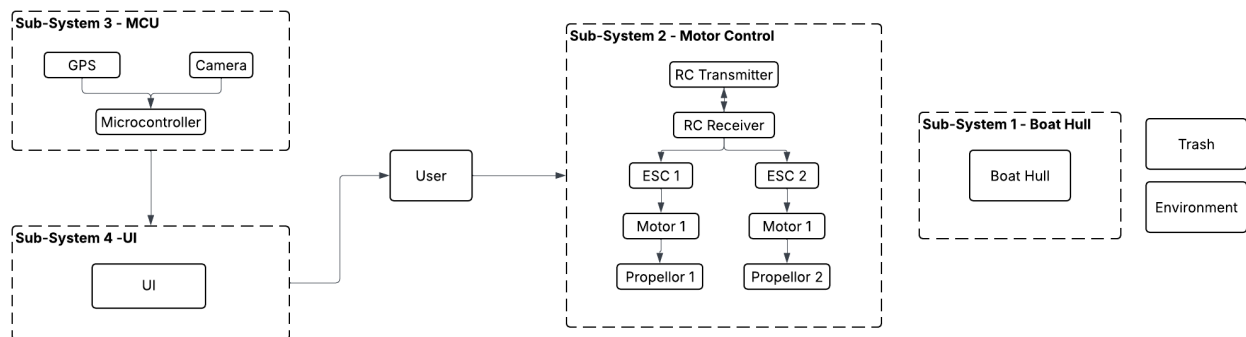


Figure 1: System Diagram Showing Boundaries

“The System” as described in this document includes two major sub-systems that both interact with the user. Sub-system one includes the GPS and Camera modules controlled by a microcontroller whose main functions are to provide feedback of the system’s state to the user via a UI. The second sub-system’s primary goal is to transform user inputs into desired movement, achieved through an RC connected to the boat’s propellers. Other important elements of the project that fall outside the controllable boundaries of the system include the trash the project wishes to collect, the system’s environment, and the Boat Hull.

Monitored & Controlled Variables

Table 2: Monitored Variables

Monitored Variables (System Inputs)	Description
Vertical Joystick Angle	This corresponds to clockwise and counterclockwise motor rotation resulting in forward and backwards TCM motion respectively, activated by moving the controller joystick forwards and backwards on the RC.
Horizontal Joystick Angle	This corresponds to right and left motor isolated activation resulting in TCM rotation, triggered by moving the controller joystick left and right on the RC.

Table 3: Controlled Variables

Controlled Variables (System Outputs)	Description
Right Motor RPM	This is the actual right motor speed that will turn the right propeller.
Left Motor RPM	This is the actual left motor speed that will turn the left propeller.
GPS Coordinates	These are the x and y coordinates of the system, that will be updated in real-time and be displayed on the UI.
Live Camera Feed	This is the live camera feed that will be updated in real-time and be displayed on the UI.

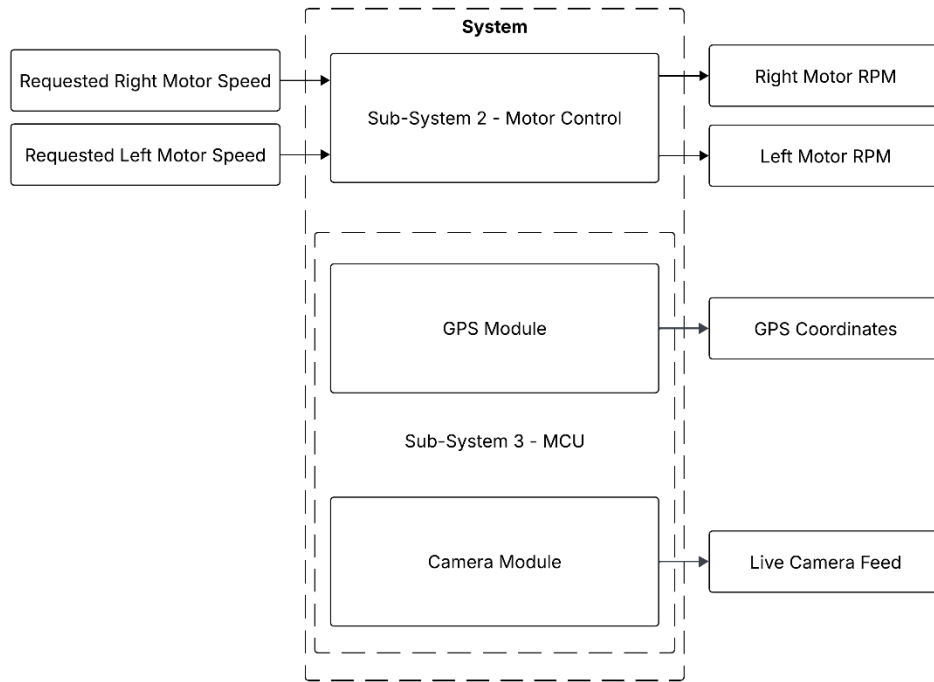


Figure 2: Context Diagram

Constants with Nominal Values & Ranges

Table 4: Constants with Nominal Values & Ranges

Constants	Nominal Values & Ranges	Units
Maximum Distance from User	The minimum of either the RC range or the Wi-Fi range of the microcontroller.	meters [m]
Maximum Speed of Motors	TBD ; based on motor specifications and testing.	RPM [revolutions/minute]
Maximum Weight on System	TBD ; based on testing.	grams [g]
Maximum Transmission Delay (RC)	Maximum time delay between transmission of RC data and receipt of said data by the receiver (on boat).	microseconds [us]
Maximum Transmission Delay (MCU)	Maximum time delay between collection of data from GPS and camera to receipt of data by the host PC (user).	microseconds [us]

Behaviour Overview

The integrated surface-water trash collecting system will securely collect trash floating on the surface of small bodies of water while operating reliably carrying all associated components and any collected trash. The TCM's movement will be remotely controlled by a user allowing forward, backward, and rotational direction changes. The system will inform users of the TCM's current location and provide live video footage of its front view to allow for optimal control decisions.

Sub-System 1: Boat Hull

Design

The boat hull design features twin-hull catamaran style boat, using two pontoons connected by two bridges. The pointed front end helps the boat cut into the water smoothly reducing drag and improving efficiency. The distance between the pontoons provides great stability, making the boat less likely to tip and allowing it to handle uneven movement in water. Inside these pontoons, there is ample enough storage for the motor and other electrical components including the propellers. The bridges allow the boat to have a flat desk area which will be used for storing components like the ESP-32, camera etc. Overall, the design focuses on stability, safety, smooth movement, and buoyancy.

Boat Hull Overview

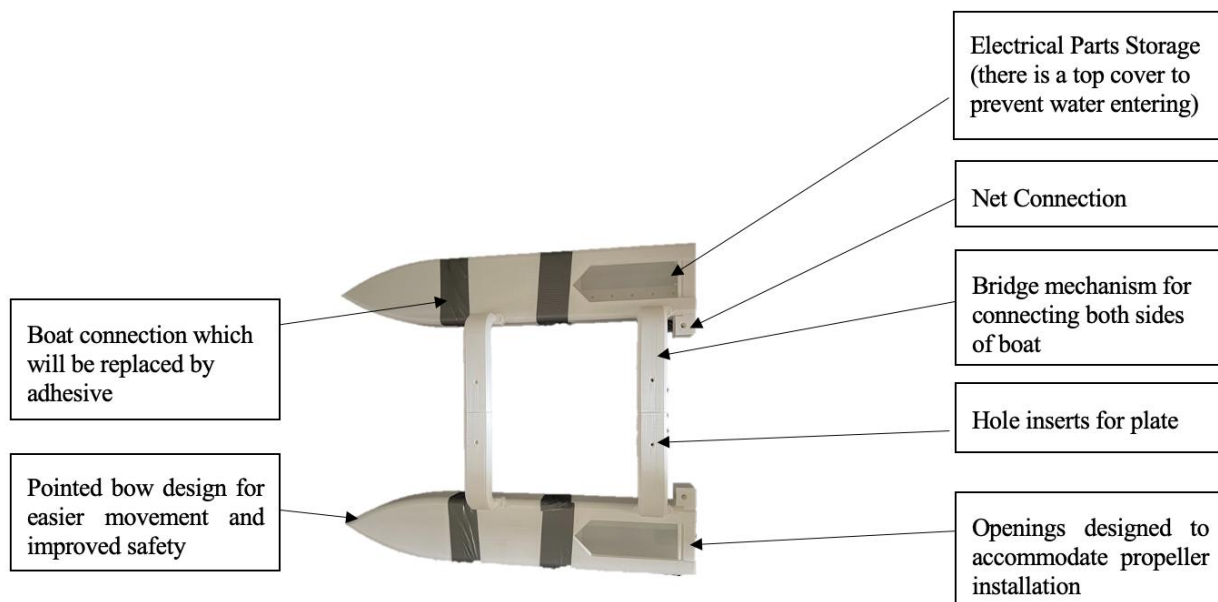
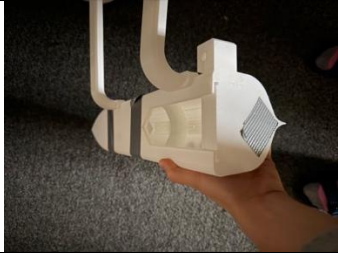
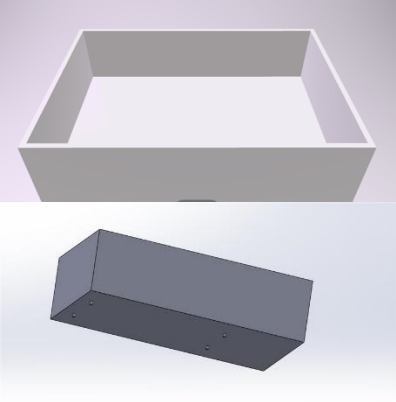
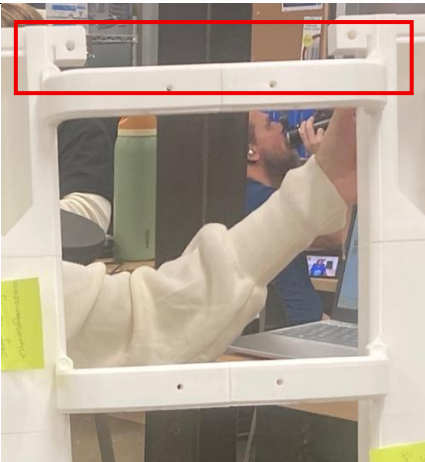


Figure 3: Boat Hull Overview Diagram

Table 5: Description of Boat Hull Components

Image	Function
	Left/Right Body: Connected via a bridging mechanism, used to keep the TCM afloat. Fins that can be seen are used to stabilize the boat and direct water flow to help with steering
	Left/Right Compartment of Boat Hull: Used to store motor, battery, wires and other electrical components that is used to control the TCM
	Shaft: Attached to the motor used to create movement and control directions. The pivot joint decreases tension exerted on the motor
	Bridging Mechanism: Used to connect the left and right side of the boat hull. Designed to be above the water level to allow surface waste to pass through as the boat hull is moving across the water.

	Plate: Attached to the bridging mechanism, used as storage compartment for the micro controller, camera, and other electrical wires/components.
	Net Connection: Used for attaching the waste net to the back of the boat hull. As the TCM moves across the body of water, surface waste will be collected into the net.

Inputs & Outputs

The inputs to the boat hull design include the payload, mass of the printed hull, safety factor, displaced water mass, and hull dimensions. These inputs are used to determine key performance outputs such as floatability, buoyancy, recovered trash capacity, boat hardware requirements (net frame, bridges, and side structures), propulsion performance, net collection rate, and operating speed based on the hull design.

Description of Behaviour

The boat hull is designed to support the necessary weight, so it can safely carry the electrical components required for propulsion. It must also generate enough thrust to overcome the drag

created by the collected trash and continue moving forward. The boat hull is designed to provide sufficient buoyancy and stability to support all onboard systems. It keeps a low center of gravity for improved stability to allow for more predictable movement. The hull is designed to keep the center of gravity low, helping it remain stable and less affected by waves or shifting loads.

Testing

Testing of the floatability and waterproof performance is performed using a kiddie pool with a radius of 1.24 meters. The pool is located indoors, allowing testing to be carried out throughout the winter. The dimensions of the pool are significantly larger than those of the boat, which ensures sufficient testing space and helps with approximate real-world operating conditions.

Through testing, it was found that the boat hull can withstand a maximum load of 2.24 kg, which is more than sufficient to support all required components.



Figure 4: Pool Visual

Sub-System 2: Motor Control

Motor Control Structural Overview & System Behaviour

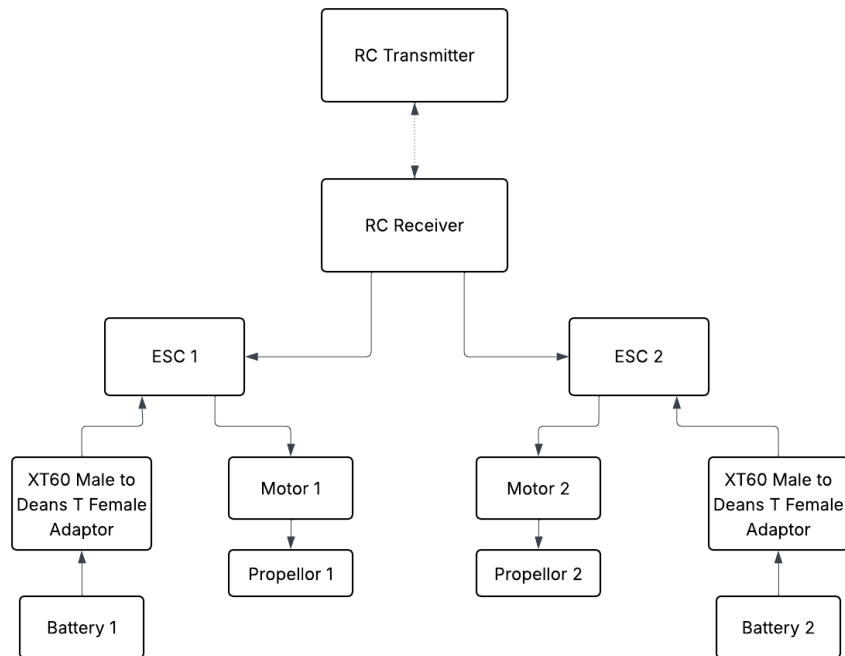


Figure 5: Motor Control System Structure

The system is comprised of an RC controller, and two sets of: ESCs, motors, propellers, XT60M to Deans T F adaptors, and battery packs. User control inputs are applied to the RC transmitter where they are communicated to ESC 1 (located on the right side of the boat) and ESC 2 (located on the left side of the boat) through RC receiver channels 2 and 5 respectively. These signals oversee controlling the rotary speed of the propellers at the stern of the boat hull through the RPM of two rotary motors. Power to the RC receiver, ESCs, and motors are provided by two battery packs and is distributed to all components through the ESC.

Inputs & Outputs

The inputs of Sub-System 2 are achieved via user input on an RC controller. Noted below are the relevant switches that the user will interact with, their input actions, the transmitter RC channels they are associated with, the receiver RC channels their output is communicated through, and what these control action achieve in the overall system.

Table 6: Functional Description of RC Controller Switches

Switch	Input	Input Channel	Output Channel(s)	Purpose
Right Joystick	Vertical Joystick Angle	N/A	Channel 2 (right motor) Channel 5 (left motor)	The vertical joystick angle controls the forward and backward movement of the TCM. A positive angle results in forward movement while a negative angle results in backward movement
Right Joystick	Horizontal Joystick Angle	Channel 1	Channel 2 (right motor) Channel 5 (left motor)	The horizontal joystick angle controls the left and right rotational movement of the TCM
Aux Switch D	Up or Down	N/A	Channel 1 Channel 2 (right motor) Channel 5 (left motor)	Aux switch D controls the power provided to the motors. Having the switch down will provide 100% power to the motors and will be referred to as "Sport Mode". Having the switch up will only provide 50% power to the motors and will be referred to as "Normal Mode".



Figure 6: RC Controller

The outputs of Sub-System 2 are the resulting motor control actions resulting from the user inputs. Noted below are the resulting motor behaviours, the RC channel control inputs are communicated through, the associated input that caused the output, and the system behavioural description.

Table 7: Functional Description of RC Receiver Outputs

Output	RC Channel	Associated Input	Description
Right Propellor Rotational Direction	Channel 2	Vertical Joystick Angle	The right propellor will rotate clockwise resulting in a forward motion when the joystick has a positive vertical angle. Similarly, the right propellor will rotate counterclockwise resulting in a backward motion when the joystick has a negative vertical angle.

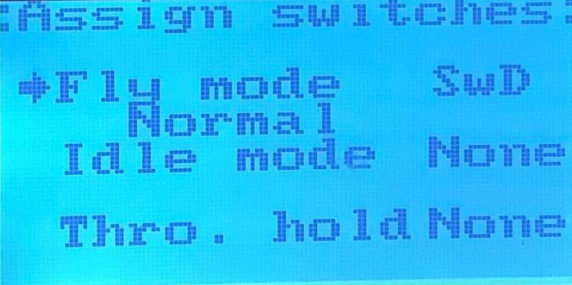
Left Propellor Rotational Direction	Channel 5	Vertical Joystick Angle	The left propellor will rotate clockwise resulting in a forward motion when the joystick has a positive vertical angle. Similarly, the left propellor will rotate counterclockwise resulting in a backward motion when the joystick has a negative vertical angle.
Isolated Propellor Movement	Channel 2 Channel 5	Horizontal Joystick Angle	When the joystick is moved horizontally to the left, the right motor will move in isolation, resulting in a counterclockwise rotational movement of the TCM (i.e. turning left). Similarly, when the joystick is moved horizontally to the right, the left motor will move in isolation, resulting in a clockwise rotational movement of the TCM (i.e. Turning right).
Motor Speed	Channel 2 Channel 5	Aux Switch D	When the switch is up the motors will operate at 50% power, resulting in a slower boat speed for garbage collection operations. When the switch is down the motors will operate at 100% power, resulting in a faster boat speed for quick boating maneuvers. Exact speeds have yet to be determined.

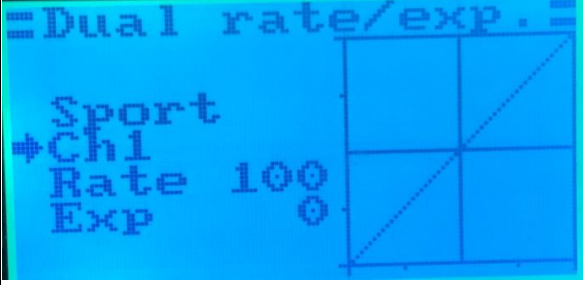
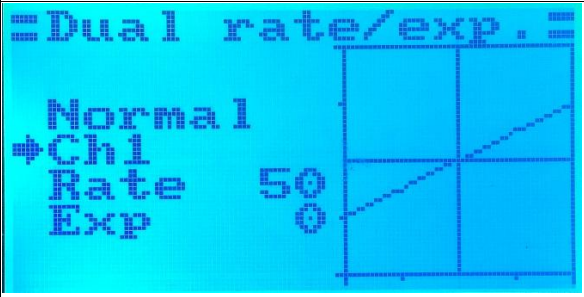
Design

Note that the Flysky FS-i6X Digital Proportional Radio Control System Instruction Manual [1] was referenced to program the RC controller.

Table 8: RC Controller Settings and Explanation

Programming Description	RC Controller
Mix 1 allows motor synchronization. The slave channel (channel 5) is programmed to mimic the master channel's (channel 2's) signals in accordance with the vertical joystick input. An offset of -45% was determined experimentally to	 <pre> =====Mixes===== ➔Mix#1 Mix is On Master ch2 Slave ch5 Pos. mix 100% Neg. mix 100% Offset -45% </pre>

properly synchronize the activation of the two motors. As a result, when forward motion is requested, both motors rotate clockwise. Likewise, when backward motion is requested, both motors rotate counterclockwise.	
Mix 2 isolates the movement of the right motor. The master channel (channel 1) corresponds to the horizontal movement of the right RC joystick. Channel 2 (corresponding to the right motor) is programmed to mimic the “positive” signals of channel 1, which are left joystick movements. Therefore, when the joystick is moved to the left, the right motor turns on synchronously. As a result, when the user moves the controller joystick to the left, the TCM turns left.	 <pre> =====Mixes===== ➔Mix#2 Mix is On Master Ch1 Slave Ch2 Pos. mix 100% Neg. mix 0% Offset 0% </pre>
Mix 3 isolates the movement of the left motor. The master channel (channel 1) corresponds to the horizontal movement of the right RC joystick. Channel 2 (corresponding to the left motor) is programmed to mimic the “negative” signals of channel 1, which are right joystick movements. Therefore, when the joystick is moved to the right, the left motor turns on synchronously. As a result, when the user moves the controller joystick to the right, the TCM turns right.	 <pre> =====Mixes===== ➔Mix#3 Mix is On Master Ch1 Slave Ch5 Pos. mix 0% Neg. mix -100% Offset 0% </pre>
This programming page allows auxiliary switch D to change operation modes of the system.	 <pre> Assign switches: ➔Fly mode SwD Normal Idle mode None Thro. hold None </pre>

Sport mode has been programmed to provide 100% of the normal operating power for the motors.	
Normal mode has been programmed to provide 50% of the normal operating power to the motors.	

Undesired Event Handling

Normal system operations of sub-system 3 are as described in the design details above. In the event of a signal disruption between the RC transmitter and receiver due to the TCM exceeding the RC communication range, power failures, water damage, wire disconnections, or any other unforeseen events, a motor failsafe shut off is triggered, automatically disconnecting power to the motors. Additionally, the RC controller will make a designated noise, informing the user of the signal disconnection and resulting motor shut off. Upon reconnection of the signal, the motors are automatically reactivated and can be controlled immediately without any manual intervention needed.

Sub-System 3: MCU

Description of Behaviour

The purpose of this subsystem is to facilitate communication between the two peripheral devices (camera and GPS) and the UI. These two peripherals interact with the surrounding environment of the boat, and existing libraries will be used to gather raw data from said peripherals.

Note that data parsing and validation will be handled within the UI subsystem, thus the only code that is included in this section will be the initialization of both peripherals. As well, we anticipate using the MicroPython machine library to handle UART communications, and the MicroPython camera driver to handle camera communications. This is the interface that Freenove recommends using when handling UART and camera communications [2].

Schematic

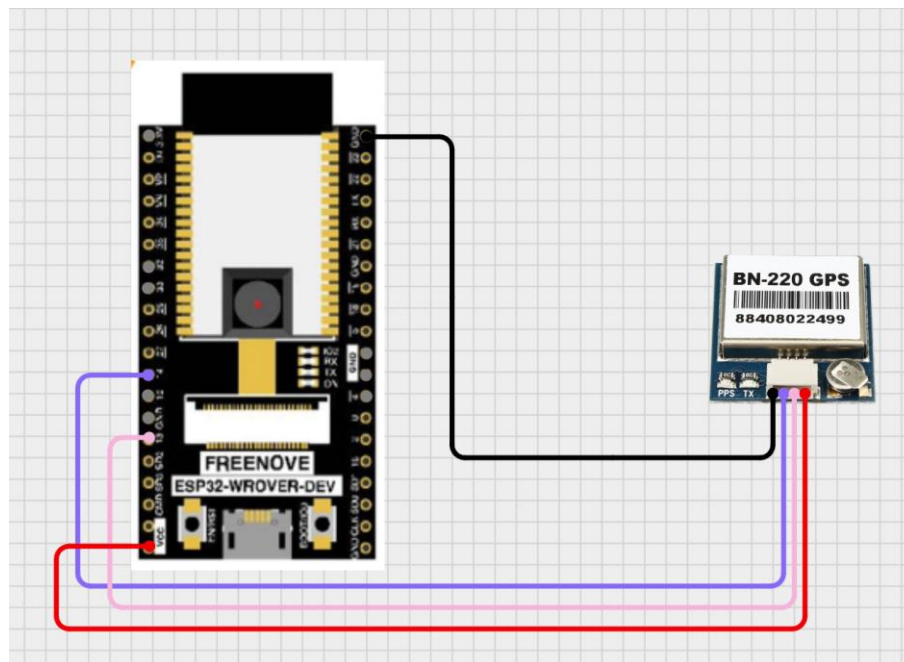


Figure 7: Sub-System 3 Electrical Schematic

Note the following:

1. According to the GPS datasheet (BN-220), the common collector voltage (VCC) range is 3.0-5.5V, with a typical value of 5.0V [3]. Thus, it is connected to 5V VCC.
2. The GPS uses a serial communication protocol to communicate with the host (MCU) [3], which includes a transmit (TX) and receive (RX) line. To the best of our knowledge, the serial communication protocol used is UART (Universal Asynchronous Receiver/Transmitter).
3. The following pins can be used for the UART bus: 0, 2, 4, 5, 9, 10, 12-19, 21-23, 25, 26, 34-36, and 39. It is recommended by the manufacturers, based off demo scripts, that pins 13 and 14 or 5 and 2 (RX and TX, respectively) can be used for UART [2]. Our group has thus chosen to use pins 13 and 14 for GPS communication.
4. The camera is connected to the MCU via a ribbon cable [2].

Inputs & Outputs

The inputs to the MCU subsystem are the local environment around the boat, and the geographic location of the boat. There are no software inputs as the purpose of this subsystem is to take environmental inputs and properly format the data to be sent to the UI.

The outputs of the MCU subsystem are as follows:

Table 9: MCU System Outputs

Output	Source	Format	Purpose
GPS Data	GPS module (MCU)	NMEA 0183	Display system position/coordinates
Camera Stream	Camera module (MCU)	Video stream	Visual observation of the system's surroundings

Note that this table matches the UI inputs table as all outputs from the MCU subsystem are to be displayed on the UI.

Software Design: Initialization

GPS

Command to initialize a UART object: UART (id, baudrate, bits, parity, rx, tx, stop, timeout) [2]

Table 10: GPS Initialization Parameters

Parameter	Description	Value	Justification
id	Serial number	1	Either 1 or 2 [2]; we only need 1 UART channel, so labelling 1 is simplest
baudrate	Baud rate	4800	Minimum value from BN-220 datasheet [3]; default baud rate for NMEA 0183 [4]
bits	Bits	8	NMEA 0183 specification [4]
parity	Odd/even checking	None	NMEA 0183 specification [4]
rx	Read pin number	13	From above schematic
tx	Write pin number	14	From above schematic
stop	Number of stop bits	1	NMEA 0183 specification [4]
timeout	Timeout period (ms)	2000	The hot start acquisition time for the GPS is 1s [3], so doubling this value ensures adequate data integrity

Camera

At the moment, camera initialization settings are to be determined. However, our group intends to start with the following command to initialize the camera:

```
camera.init(0, format=camera.JPEG, fb_location=camera.PSRAM) [2]
```

These settings ensure that the camera output is in JPEG format and enables the use of PSRAM, which will allow the camera to capture higher quality images. However, these parameters are subject to change, and additional parameters may be changed from their default values if our team finds that their default settings are insufficient for the user experience.

Sub-System 4: UI

Inputs & Outputs

Table 11: UI System Inputs

Input	Source	Format	Purpose
GPS Data	GPS module (MCU)	NMEA 0183	Display system position/coordinates
Camera Stream	Camera module (MCU)	Video stream	Visual observation of the system's surroundings

Table 12: UI System Outputs

Output	Destination	Purpose
Map Display	User	Reporting location of the system
Live Video Feed	User	Trash identification/situational awareness

Description of Behaviour

Upon startup, the UI initializes map view and video display components. When valid GPS data is received from the GPS module, the UI updates the map marker to reflect the system's current position. If GPS data becomes unavailable or invalid, the UI retains the last known position and displays a warning indicator. The live camera feed is displayed continuously. Loss of the camera feed results in a visual alert to the user while maintaining the map display.

Software Design

The UI software is decomposed into display and data-handling modules to ensure clear separation between data acquisition, validation, and visualization. As the UI is observational only, no control logic is implemented.

Module Guide

Table 13: UI System Module Guide

Module	Responsibility	Module Secrets (Information Hiding)
Data Receiver	Receives GPS module and camera module data from the MCU.	Communication protocol used to acquire data (Wi-Fi), message framing, and data acquisition mechanisms.
Data Validator	Validates the eligibility and freshness of the data.	Timeout thresholds to make sure that the data presented is the newest set coming from the MCU.
Map Renderer	Displays GPS data as a marker on a map.	Map libraries used, map projection.
Video Renderer	Displays camera feed.	Video buffering minimization strategy, frame decoding, and rendering timing used to display the most recent camera frame.
UI Controller	Manage UI operational states (initializing, normal, degraded). Coordinate updates to map and video displays. Trigger user-visible warnings when data is invalid.	Internal UI state management, panel layout constraint handling, and coordination logic for updating renderer modules based on system state and data validity.

Module Decomposition

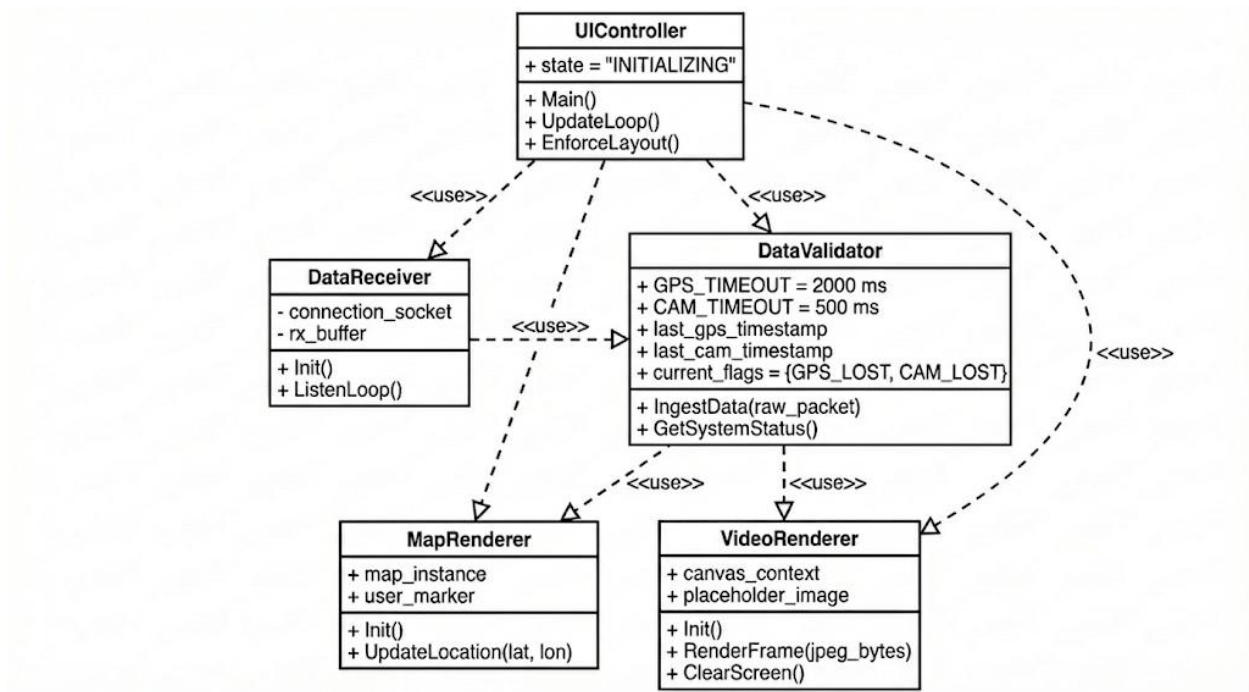


Figure 8: UML Class Diagram

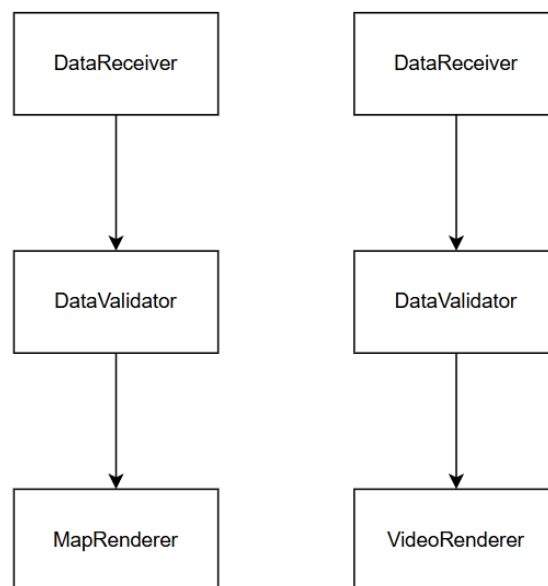


Figure 9: GPS Module (Left Column) & Camera Module (Right Column)

In Figure 7, the **UIController** will be used for both the GPS and Camera modules.

Module Internal Design (MID)

The MID has been described for each module in pseudocode below, matching the dependencies in the UML Class Diagram.

Class DataReceiver:

```
Function Init():
    //IP address and port number instantiated
    connection_socket = OpenSocket("192.168.4.1", 80)
    rx_buffer = EmptyBuffer()
Function ListenLoop():
    While True:
        // reads available bytes from network
        raw_bytes = connection_socket.read()
        IF raw_bytes is not Empty:
            // accumulates bytes in the buffer
            rx_buffer.Append(raw_bytes)
            // processes buffer while it contains full
packets
            While rx_buffer.HasCompletePacket():
                //passes raw data directly to the DataValidator
class
                packet = rx_buffer.PopPacket()
                DataValidator.IngestData(packet)
        ELSE:
            Wait(10ms)
```

Class DataValidator:

```
Constant GPS_TIMEOUT = 200 ms
Constant CAM_TIMEOUT = 50 ms
Property last_gps_timestamp
Property last_cam_timestamp
Property current_flags = {GPS_LOST, CAM_LOST}
// Called by Data Receiver
Function IngestData(raw_packet):
    IF Checksum(raw_packet) is VALID:
        IF raw_packet.type == "GPS":
            last_gps_timestamp = Now()
            current_flags.GPS_LOST = False
            StoreValidGPS(raw_packet.payload)
        ELSE IF raw_packet.type == "VIDEO":
            last_cam_timestamp = Now()
            current_flags.CAM_LOST = False
            StoreValidFrame(raw_packet.payload)
    ELSE:
        Log("Corrupt Packet Dropped")
// Called by UI Controller to check system health
Function GetSystemStatus():
    // Check for timeouts (Freshness)
    IF (Now() - last_gps_timestamp) > GPS_TIMEOUT:
        current_flags.GPS_LOST = True
    IF (Now() - last_cam_timestamp) > CAM_TIMEOUT:
        current_flags.CAM_LOST = True
```

```
RETURN current_flags, GetStoredGPS(), GetStoredFrame()
```

Class MapRenderer:

```
Property map_instance  
Property user_marker  
Function Init():  
    map_instance = LoadMapLibrary("Leaflet")  
    user_marker = CreateMarker(0, 0)  
Function UpdateLocation(lat, lon):  
    // Convert coords to map projection pixels  
    position = GeoToPixel(lat, lon)  
    // Move icon  
    user_marker.SetPosition(position)  
    // Auto-center view  
    map_instance.PanTo(position)
```

Class VideoRenderer:

```
Property canvas_context  
Property placeholder_image  
Function Init():  
    canvas_context = GetHTMLCanvas("video-feed")  
Function RenderFrame(jpeg_bytes):  
    image_object = DecodeJPEG(jpeg_bytes)  
    canvas_context.DrawImage(image_object)  
Function ClearScreen():  
    // Called when signal is lost  
    canvas_context.DrawImage(placeholder_image)
```

```
canvas_context.DrawText("NO SIGNAL", Center)
```

Class UIController:

```
Property state = "INITIALIZING"
```

```
Function Main():
```

```
    DataReceiver.Init()
```

```
    MapRenderer.Init()
```

```
    VideoRenderer.Init()
```

```
    StartThread(DataReceiver.ListenLoop)
```

```
    StartTimer(UpdateLoop, 30fps)
```

```
Function UpdateLoop():
```

```
    // gets validated state
```

```
    flags, gps_data, frame_data =  
    DataValidator.GetSystemStatus()
```

```
    // handles undesirable behaviour (signal loss)
```

```
    IF flags.CAM_LOST OR flags.GPS_LOST:
```

```
        TriggerWarning("Connection Unstable")
```

```
    IF flags.CAM_LOST:
```

```
        VideoRenderer.ClearScreen()
```

```
    IF flags.GPS_LOST:
```

```
        // greys out map or show last known location
```

```
        ShowToast("GPS Signal Lost")
```

```
    // normal operation
```

```
    IF NOT flags.CAM_LOST:
```

```
        VideoRenderer.RenderFrame(frame_data)
```

```
    IF NOT flags.GPS_LOST:
```

```

        MapRenderer.UpdateLocation(gps_data.lat, gps_data.lon)
    // enforce layout constraints
    EnforceLayout()
Function EnforceLayout():
    // Ensure map and video don't overlap on small screens
    screen_width = GetWindowWidth()
    IF screen_width < 600px:
        SetStyle("map-panel", "width: 100%; height: 50%")
        SetStyle("video-panel", "width: 100%; height: 50%")
    ELSE:
        SetStyle("map-panel", "width: 30%; height: 100%")
        SetStyle("video-panel", "width: 70%; height: 100%")

```

Undesired Event Handling

Table 14: Undesired Event Handling

ID	Event	Response	Rationale
E1	UI loses connection to the GPS and camera modules	Warning on the UI to inform users of issues with data retrieval. The UI is able to identify this issue via the timeout feature.	This makes the user aware that there is an issue in the Wi-Fi connection, and they can attempt to reconnect with the modules.
E2	Net rips/detaches from Boat Hull	User retrieves the TCM and manually switches the net out and recollect the “leaked waste”	Switching out the net and recollecting the waste ensures the waste collection process is completed and prevents the system from producing excess waste.

Timing Constraints

The following requirements define the performance expectations of our surface-water trash collector, specifying how quickly the system must detect, collect, and dispose of floating debris in West Pond.

Table 15: Performance Timing Requirements

RID	Description	Likelihood of Changing
R8	Detection Response Time: Using camera visualization, the user should be able to detect a standard floating object approaching the TCM within TBD seconds.	Low
R9	Collection Cycle Time Per Item: The TCM should complete a full collection time under normal conditions (user detect → align → capture → store) for a properly sized item in less than TBD .	Medium
R10	Battery Efficiency Cycle: The system shall operate continuously for at least TBD hours before it needs to be recharged.	Low

R11	Alignment Time: The TCM should properly align the net to intercept an object within TBD sec from TBD meters away from the net.	Medium
R12	Direction Change Time: The TCM should be able to change direction by TBD degree angle in TBD seconds.	Low
R13	Thrust to Drag Response Time: The propulsion mechanism shall generate sufficient thrust to overcome drag and provide forward motion within TBD seconds of motor activation from rest. <i>Rationale:</i> This ensures that the system can overcome water and collected trash resistance quickly, preventing stall at low speeds.	Low
R14	Motor Response Time: Each of the motors should reach TBD of the user input speed within TBD seconds of receiving the control signal.	Low
R15	GPS Response Time: The delay between receiving GPS data from the BN-220 and the ESP32 main control loop processing shall be $\leq$ TBD	Low

References

- [1] Flysky RC model technology co., ltd, "Flysky FS-i6X Digital Proportional Radio Control System Instruction Manual," Flysky RC model technology co., ltd, 2016.
- [2] Freenove, "Freenove ESP32-WROVER Board Python Tutorial," [Online]. Available: <https://docs.freenove.com/projects/fnk0060/en/latest/fnk0060/codes/Python.html>.
- [3] Beitian Co., Limited, "BN-220 GPS Module +Antenna Datasheet," October 2015. [Online]. Available: <https://files.banggood.com/2016/11/BN-220%20GPS+Antenna%20datasheet.pdf>.
- [4] Active Research Limited, "The NMEA 0183 Information sheet," [Online]. Available: <https://actisense.com/wp-content/uploads/2020/01/NMEA-0183-Information-sheet-issue-4-1-1.pdf>.